

“ THEN WAS ALL OVER ENGLAND
SUCH A TOKEN SEEN AS NO MAN
EVER SAW BEFORE... ”

From *The Anglo-Saxon Chronicle*, describing the comet of 1066

IN THE 11TH CENTURY, Chinese mathematician **Jia Xian** described a method of **calculating square and cubic roots** using numbers arranged in rows. Each row contained one more number than the row above it, to form a triangle in which each number is the sum of the two directly above it. Known as the **Jia Xian triangle**, it is also often referred to in the West as Pascal's triangle, after French mathematician Blaise Pascal, who described it 600 years later.

In 1054, the massive explosion of a **supernova** (which formed what we now know as the Crab

22 THE NUMBER OF MONTHS THAT THE **SUPERNOVA** WAS **VISIBLE** FROM **1054 TO 1055**

Nebula) was visible from Earth. It was **observed by Arabic and Chinese astronomers**, who described it as a “guest star,” but its significance was not realized by observers in Europe.

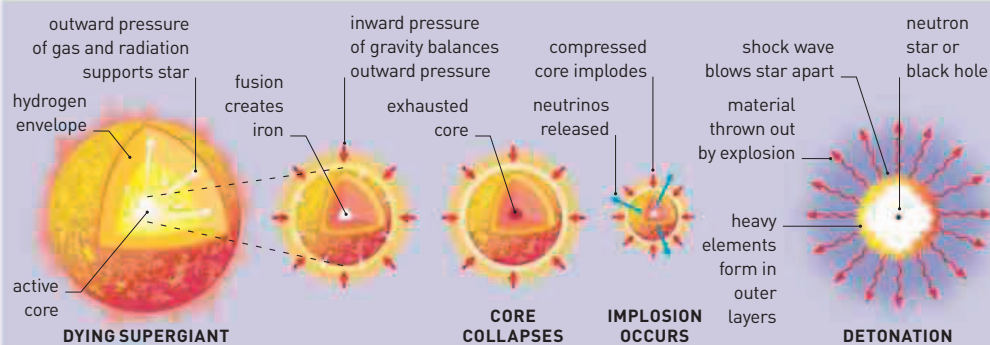
In 1066, the comet now called **Halley's Comet** made one of its regular 76-year periodic appearances and was described

by European astronomers.

Astrologers viewed the comet as an omen, and found it especially significant in the year of the Norman invasion of England.

The Bayeux tapestry

This embroidered record of the events surrounding the Battle of Hastings in 1066 shows the appearance of Halley's Comet.



FORMATION OF A SUPERNOVA

A supernova is an explosion of a massive supergiant star at the last stage of its life. Over a long period of time, a star builds up a core of iron, which eventually collapses in on itself as the star runs out of fuel for fusion. This results in an implosion that rapidly reheats the star and restarts the process of fusion.

Subatomic particles called neutrinos are released as implosion occurs. Now out of control, the star explodes with a huge amount of energy—billions of times more than the Sun, which is also a star—shining brighter than other stars and scattering debris in all directions, over vast distances.



c.1050 Jia Xian describes the Jia Xian triangle, later known as the Pascal triangle

1054 Chinese and Arab astronomers observe the supernova that forms the Crab Nebula

1066 The comet later known as Halley's Comet is sighted